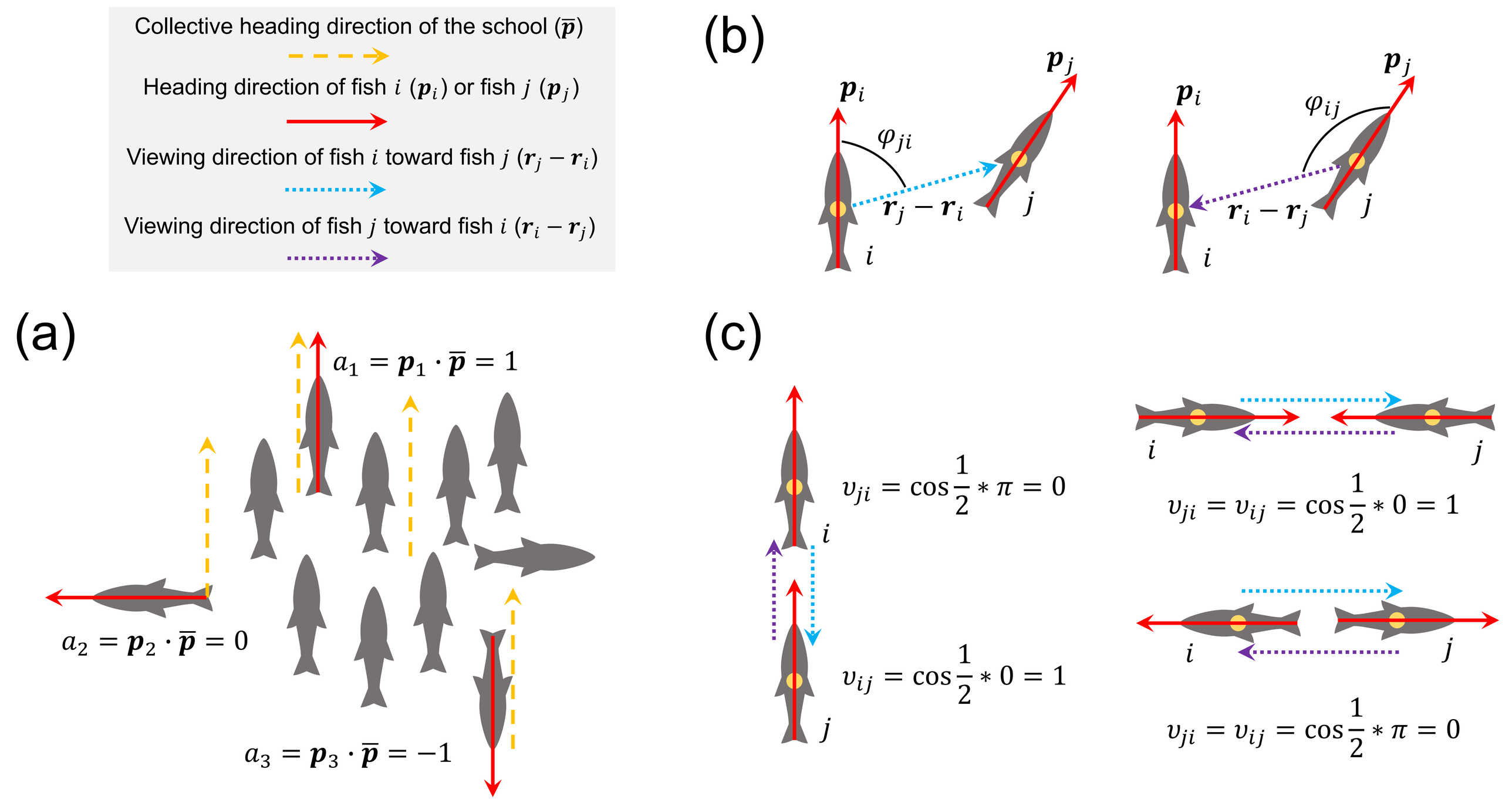
Collective heading direction of the school ($\overline{\boldsymbol{p}}$)
Heading direction of fish i ($\boldsymbol{p}_i$) or fish j ($\boldsymbol{p}_j$)
Viewing direction of fish i toward fish j ($\boldsymbol{r}_j - \boldsymbol{r}_i$)
Viewing direction of fish j toward fish i ($\boldsymbol{r}_i - \boldsymbol{r}_j$)
(a)
$a_1 = \boldsymbol{p}_1 \cdot \overline{\boldsymbol{p}} = 1$
$a_2 = \boldsymbol{p}_2 \cdot \overline{\boldsymbol{p}} = 0$
$a_3 = \boldsymbol{p}_3 \cdot \overline{\boldsymbol{p}} = -1$
(b)
$\boldsymbol{p}_i$
$\boldsymbol{p}_j$
φ_{ji}
$\boldsymbol{r}_j - \boldsymbol{r}_i$
i
j
φ_{ij}
$\boldsymbol{r}_i - \boldsymbol{r}_j$
(c)
$v_{ji} = \cos\frac{1}{2} * \pi = 0$
$v_{ij} = \cos\frac{1}{2} * 0 = 1$
$v_{ji} = v_{ij} = \cos\frac{1}{2} * 0 = 1$
$v_{ji} = v_{ij} = \cos\frac{1}{2} * \pi = 0$